

Power Supply

Before turning on, set PR1 cw and PR2 ccw with only A board plugged-in.

Set PR2 for $+12V \pm 0.1V$

Set PR1 for $-9V \pm 0.1V$

Check PR2 again and reset if necessary

N.B. -9V rail is parasitic on the +12V supply.

Plug in B and C cards and recheck +12V and -9V rails.

These should not have altered. If they have, suspect either B or C board or poor regulation on A board. Plug-in keyboard as extra load. +12 and -9 Volt rails should not have moved more than $\pm 0.1V$ measured always on the A board. (C board voltage may be 20 mV lower)

Monitor the rails on the scope (5mV/cm). Look for hum and noise. These should be much less than 5mV. Noise, if present, will probably be due to the zener diode.

Power supplies should stabilise on full load for specified fluctuations.

110/240 nominal : $+10\%$ or -23%

110 nominal : 100V-135V for 125 sufficient

240 V nominal: 200V-260V

Hum can also be checked by the following:

Check Output Filters on 10

Output Controls on 10.0.10.10

Patchboard vacant

Monitor (5mV/cm) on signal output

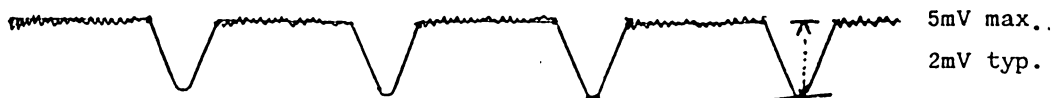
Load unit with keyboard

100 Hz ripple as shown.

Noise can be seen which should never exceed 10mV p.p. Even with noise level up. With noise level down, the noise on this waveform should be quite low. Zener diode noise will be very obvious.

N.B. A2 normally gets hot, dodgy D1 gives large 100Hz ripple on Output.

Always clean edge-connectors and check clean presets on A board.



Output Channels

Set Output Channel Filters to 5.

Set Output Level row to 0.0.10.0 and connect traces of scope (5mV/cm) to both signal outputs.

Connect full level ramp signal at 261 Hz from Oscillator 1 to both Outputs. Trim PR4 and PR5 for minimum signal breakthrough. (Maximum 5mv p.p)
Replace presets if adjustment is erratic.

Reset Output Channel row controls to 5.0.10.5. Output levels should be around 1V p.p and should be within 20% of each other. Check signal tracking in the same manner for Output settings 4 and 6 as well. If one level is consistently high, pad out resistor across its control pot (R305 = 56K)

Successful tracking will depend on the characteristics of the pots and the correct matching of the FETs Q14 & Q18.

Connect the joystick row via two shorting pins to both VC level inputs. Swing stick up and down to check for 20% level tracking tolerance. (Level pots should be at zero). Inaccuracies in stick tracking may be adjusted by altering the value of R302 or R304 (10K) which are mounted under the patchboard rear.

Check correct operation of Pan controls, which only affect the Signal Outputs. Check also the Output Filters. Use noise as a good source.

Check for max. outputs on the Signal Output socket of at least 2.5V p.p before the onset of clipping. Level controls should be at about 6. Check headphone signal is good. Check for instability of Output Amps by driving hard with low frequency. (Oscillator 3 square wave). The level control of the channel NOT being driven should be on zero. Monitor the two signal outputs. The undriven channel will show breakthrough. Plug in headphones and check for oscillation. Reduce drive to Output Channel 2. Keep checking for oscillation. Then repeat for other Output Channel.

N.B. DC offset across loudspeaker is $1.8V \pm 10\%$.

Reverberation

Set PR3 fully clockwise before operating. Feed large amplitude tone bursts into reverb. Adjust PR3 so that the output is dry at Mix setting of zero and reverberation progressively increases from setting of 2.

Maximum dryness on the Mix control should be no worse than that obtained with the VC Mix input row on the patchboard grounded.

Check voltage control of Mix with the joystick.

Hold input amplitude constant and sweep frequency of tone. Check for rattles, buzzes or abnormal resonances in the spring. Check spring is correct way round. If Mix control is wrong change Q10.

N.B. Reverb amplifier should clip symmetrically.

Envelope Shaper

Put scope on 2V/cm and connect to Trapezoid output. With envelope controls at 0.0.0.0. Adjust PR12 so that frequency of trapezoid oscillation is about 90% of the maximum possible frequency. Connect Filter output on 261 Hz (Frequency setting 5) with maximum output to Envelope Input. Monitor Envelope output. Set PR13 clockwise.

Set envelope controls 0.0.0.10. Adjust PR7 for minimum signal. Adjust PR13 until signal reappears, then back off slightly. The final envelope shaper signal breakthrough should be 5 mV or less p.p. With the Attack button pressed the output signal should be greater or the same as the input signal.

Set controls 0.10.0.10. this gives maximum On time. Press Attack button which should recycle the envelope shaper and lamp should come On, staying on for about 5 seconds. (absolute minimum 4 seconds). This corresponds to an automatic recycle On time of 2.5 seconds which is the minimum spec. (On time is always shorter if recycled automatically). On time is controlled by R148 in conjunction with Q45. f ,

Set controls 0.0.10.5. Press Attack button and time the Decay time. This should exceed 15 seconds, and the envelope shaper must automatically recycle at the end of the decay.

Set controls 0.0.0.7 to 10. Envelope shaper must not recycle with the Off time on 10, so check it doesn't on 7 or so, which will indicate the behaviour on 10. Check input breakthrough at the same time, and adjust PR7 again if necessary.

On and Off times are controlled by R148 (range 10-18K). If On time is infinite at 10, reduce R148, if Off time infinite, increase R148. The value of Vp for Q45 (3.5 - 4.0 nominal) is critical. Failure here upsets recycling. Also leaky C48 or dodgy envelope trigger components .

Filter/Oscillator

With Response on 10, level full and Frequency on 5, check for oscillation. About 1.5V p.p should be obtained. The shape should be reasonably sinusoidal especially when only just oscillating (on Response control).

Distortion should be eliminated either by diode bridging or by replacement of offending diode. Bad diodes can be detected by applying the tip of a soldering iron quickly to the diode body. Allow enough time for the diode to cool down before moving on to the next one.

Response control should make the filter oscillate only above 5 for all frequency settings. If this does not occur correctly, adjust R90 or suspect wrong Vp FET. Frequency range on the Frequency knob should be roughly 10Hz to 12kHz. If oscillations stop at one extremity of Frequency pot increase R75. Adjust PR6 for oscillation frequency of 261Hz with F on 5. Noise level of Filter output signal with Response set below point of Osc should be below 5mV p.p.

Check correct filtering using square wave input with swept F.

Intermittent noise: Try soldering iron tip on diodes.
R288 = 1 megohm stops Filter latching-up.

Ring Modulator

Use filter sinewave output, F on 5. Maximum response, maximum level into both inputs of ring modulator. Ring Modulator level maximum. Check output is double frequency of input, and at least as big. Remove pin to Input B. Drive input A and adjust presets for minimum breakthrough. (PR9 - second harmonic rejection, PR10 - fundamental rejection).

Drive Input B by removing pin to Input A and reinserting pin for input B. Trim presets for rejection. (PR8 - second harmonic, PR11 for fundamental). This one should NOT drift. Input A drifts quite a bit and should ideally be adjusted with the back on the machine after it has fully warmed-up. If this is not possible ensure that Input B is used for the continuous signal to minimise breakthrough.

With an input level of 1.5V p.p up to 10mV p.p maximum is allowed for breakthrough. This gives about 60 dB rejection.

If higher harmonics are a problem, then filter signal must be too impure. Try backing off response, or use an external signal generator. 1.5V p.p at 261 Hz.

If reasonable rejection is unobtainable, change matched pair BC169C or TAB101 where appropriate. 2C746 used on later models. Do not bridge with resistors. Replace presets if adjustment is erratic.

Oscillator 1

The first operation is to adjust the shape of the sinewave, since the adjustment affects frequency. Using the vernier control and PR15 if necessary, obtain an output of 400Hz. Turn the ramp down and adjust the sine shape control for the best sine shape. Trim PR17 to balance out the spike on the peak of the sinewave. This may best be seen by expanding the X and Y inputs on the scope.



The best position is shown, where the small spike is triangular and balanced. A small thin spike may also appear but is not important.

Test the main shape control. A good sinewave should be achieved on 5. ($\pm \frac{1}{2}$ division). If further out, R197 or R201 may need to be adjusted.

Set PR16 halfway and leave it alone. Set frequency vernier dial to 6 and adjust PR15 for 261Hz. Rotate vernier to 8 and check for 2088Hz. Absolute maximum error ± 1 division. For fudge procedure, see Osc 2 writeup.

Sine Output should be around 1.5V p.p

Ramp Output should be around 1.5V p.p

N.B. Ramp shape is not controllable.

Pitch characteristic is 0.32 Volts/Octave.

Oscillator 2

Set square level to zero, ramp at maximum. Set PR20 halfway. Vernier on 6. Trim PR19 for 261Hz. Increase vernier to 8. Check frequency is about 2088Hz. If beyond spec then R291 SOT will have to be added in the range 8-22K. Check that 10 on vernier dial gives at least 10kHz. If vernier dial at 8 gives less than 2088Hz then short circuit R291 and R213 will have to be reduced. These should be 2% resistors. Shape controls should operate identically to those on Osc 3 (see next page) except that levels are reduced, and the two outputs are summed into the patchboard on Mark 11 machines. Remember to turn down unwanted signal. Resistors R226 and R228 are the relevant ones to set the shape. Pulse level should be 2V p.p. Ramp level should be 3.5V. N.B. Osc2 and 3 waveforms are not symmetrical with respect to zero volts.

Tracking

Mix Osc 1 and Osc 2 ramp outputs into Output Channel 1. Use PR20 only to trim the tracking of Osc 1 and Osc 2. Using the joystick on max with range control to limit frequency range. Set both verniers to 6 and tune for zero beats. Advance the joystick range control which will increase both frequencies until the beats begin. Trim PR20 to zero beats, and remember which way you turned it. (N.B. The preset is a very fine adjustment, and when the tracking is nearly right it will be necessary to make very very small movements). Having trimmed PR20, turn the range control back to zero, readjust one of the verniers near 6 to give zero beats and then advance range control of joystick to where it was before. If beats are faster you know you turned the preset the wrong way. Repeat procedure until tracking goes to about 2kHz. Ideally a test 'back' should now be fitted and a final adjustment made after another 10 minutes have elapsed.

Faults

Tracking from 261Hz to about 2kHz should be achieved with a maximum of 3 beats per second on the way. A curved Voltage/Frequency response may occur where zero beats are produced in more than one place. This is due to a mismatch in the matched pair transistors (Q74 & Q75 for Osc 2). Fit a new transistor here or in Osc 1.

N.B. Always use specially matched 2K7 pins for tracking tests.

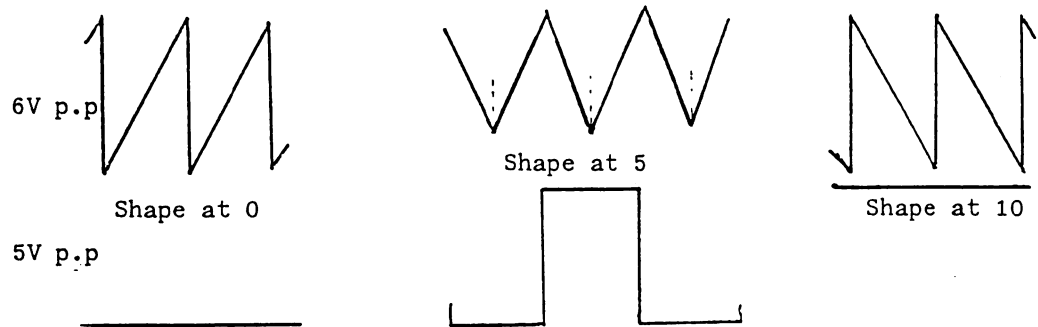
Oscillator 3

Check frequency control pot for even change in frequency.

Adjust PR21 so that when the vernier is at 8 the frequency is 63Hz.

Note: Vernier 10 should give 500Hz. Period at 0 should be greater than 20 sec.

Adjust values of R261 & R262 with pad resistors to achieve correct shapes of waveforms as shown below:

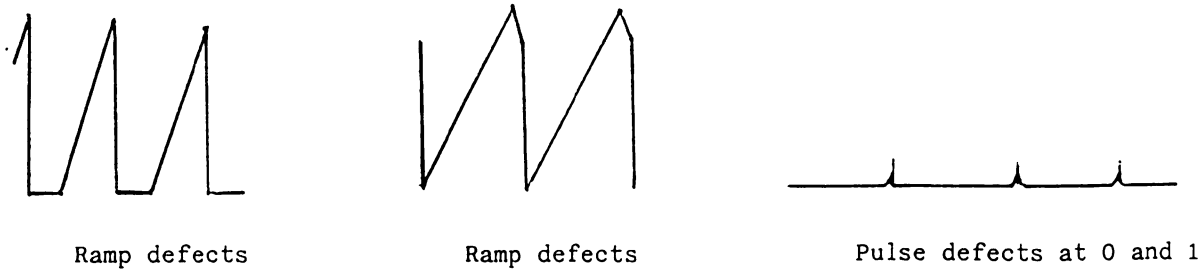


Triangle and square wave should occur on 5. Maximum of $\frac{1}{2}$ division error allowed.

Check Voltage control input.

Faults

Osc 3 breaks through to Osc 2 and to a lesser extent Osc 1. Effect dependent on Osc 3 shape control setting. Triangle was 'furry', with 10mHz oscillations visible all over C board. Strongest near Q101. Cause: instability in Q99. Q100, Q101. Cure, C70 330pF change to 1nF.



Noise Generator & Meter Amplifier

Colour control on 5, max output. Select a noise diode which gives wideband noise and can be adjusted to give 3V p.p with PR22 roughly midway (to give some later adjustment margin).

Lumpy or crackly sounding noise diodes should be rejected.

Meter

Check left hand zero of meter with machine off or switched to Signals.

Switch to Control Voltages and adjust PR14 for accurate centre zero.

Switch the meter switch a few times and recheck zero again.

Patch the joystick to Meter for swings in both + and - directions.

Check for needle sticking at either end of travel. Switch to Signals and test with Osc 2 sawtooth at max level. Vary level to check proportionality.

Faults

Watch out for the resistor R170 on the C board, which often touches the -ve end of C76, and has been known to produce curious failures on soak test.